

Computational Modeling of Bound and Pseudo-Bound States in the Yukawa Potential

Advisor: Dr. Jianyi Jay Wang

Charles J. Mikolajczak

University of Massachusetts – Dartmouth
Department of Physics

05/06/2026



- ① Background
- ② Goals
- ③ Methods
- ④ Results & Discussion
- ⑤ Moving Forward
- ⑥ AI Implementation
- ⑦ References

1 Background

2 Goals

3 Methods

4 Results & Discussion

5 Moving Forward

6 AI Implementation

7 References

What is the Yukawa potential?

- Hideki Yukawa (1907-1981)



Hideki Yukawa (1907-1981)

What is the Yukawa potential?

- Hideki Yukawa (1907-1981)
- Early description of the strong nuclear force, introducing a massive particle now known as a meson



Hideki Yukawa (1907-1981)

What is the Yukawa potential?

- Hideki Yukawa (1907-1981)
- Early description of the strong nuclear force, introducing a massive particle now known as a meson

- $$V(r) = \begin{cases} -V_0, & r \leq a \\ \frac{e^{-\lambda r}}{r}, & r > a \end{cases}$$



Hideki Yukawa (1907-1981)

What is the Yukawa potential?

- Hideki Yukawa (1907-1981)
- Early description of the strong nuclear force, introducing a massive particle now known as a meson
- $$V(r) = \begin{cases} -V_0, & r \leq a \\ \frac{e^{-\lambda r}}{r}, & r > a \end{cases}$$
- Lays foundation for modern nuclear and atomic physics



Hideki Yukawa (1907-1981)

1 Background

2 Goals

3 Methods

4 Results & Discussion

5 Moving Forward

6 AI Implementation

7 References

Why is this important?

- Provides a simple computational model for interactions between nucleons

Why is this important?

- Provides a simple computational model for interactions between nucleons
- Allows for visualization for bound and pseudo-bound states

Why is this important?

- Provides a simple computational model for interactions between nucleons
- Allows for visualization for bound and pseudo-bound states
- Provides calculation for systems beyond the simple Hydrogen model

Goals

- Build a solver in Python based on `hydrogen.py`

Goals

- Build a solver in Python based on `hydrogen.py`
 - Solve for bound and pseudo-bound states with associated eigenenergies

Goals

- Build a solver in Python based on `hydrogen.py`
 - Solve for bound and pseudo-bound states with associated eigenenergies
 - Produce plots for potential, reduced radial wave function $u_{nl}(r)$, and eigenenergies

Goals

- Build a solver in Python based on `hydrogen.py`
 - Solve for bound and pseudo-bound states with associated eigenenergies
 - Produce plots for potential, reduced radial wave function $u_{nl}(r)$, and eigenenergies
 - Compare results with differing strength and range parameters

1 Background

2 Goals

3 Methods

4 Results & Discussion

5 Moving Forward

6 AI Implementation

7 References

Mathematics

- Radial Schrödinger Equation:

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[V(r) + \frac{\hbar^2}{2m} \frac{\ell(\ell+1)}{r^2} \right] u = Eu$$

Mathematics

- Radial Schrödinger Equation:

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[V(r) + \frac{\hbar^2}{2m} \frac{\ell(\ell+1)}{r^2} \right] u = Eu$$

- Numerov's Method:

$$u_{i+1} = \left(1 + \frac{h^2}{12} f_{i+1} \right)^{-1} \left[2 \left(1 - \frac{5h^2}{12} f_i \right) u_i - \left(1 + \frac{h^2}{6} f_{i-1} \right) u_{i-1} \right]$$

Mathematics

- Radial Schrödinger Equation:

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[V(r) + \frac{\hbar^2}{2m} \frac{\ell(\ell+1)}{r^2} \right] u = Eu$$

- Numerov's Method:

$$u_{i+1} = \left(1 + \frac{h^2}{12} f_{i+1} \right)^{-1} \left[2 \left(1 - \frac{5h^2}{12} f_i \right) u_i - \left(1 + \frac{h^2}{6} f_{i-1} \right) u_{i-1} \right]$$

- Shooting Method:

$$f(E) = u_m^\downarrow w_m^\uparrow - u_m^\uparrow w_m^\downarrow, \quad w = u' \implies w' = u''$$

Mathematics

- How can we implement these equations in a simulation?

Mathematics

- How can we implement these equations in a simulation?
 - Introducing... Python!

Computation

- Use `hydrogen.py` source code as foundation

Computation

- Use `hydrogen.py` source code as foundation
 - $V(r) = 1/r \rightarrow V(r) = e^{-\lambda r}/r$

Computation

- Use `hydrogen.py` source code as foundation
 - $V(r) = 1/r \rightarrow V(r) = e^{-\lambda r}/r$
- Packages: NumPy, Matplotlib, SciPy

Computation

- Use `hydrogen.py` source code as foundation
 - $V(r) = 1/r \rightarrow V(r) = e^{-\lambda r}/r$
- Packages: NumPy, Matplotlib, SciPy
- Define functions for potentials, Numerov method, and shooting method

Computation

- Use `hydrogen.py` source code as foundation
 - $V(r) = 1/r \rightarrow V(r) = e^{-\lambda r}/r$
- Packages: NumPy, Matplotlib, SciPy
- Define functions for potentials, Numerov method, and shooting method
- Calculate energy and approximate wave function in a `for` loop over initialized ℓ -values

Computation

- Use `hydrogen.py` source code as foundation
 - $V(r) = 1/r \rightarrow V(r) = e^{-\lambda r}/r$
- Packages: NumPy, Matplotlib, SciPy
- Define functions for potentials, Numerov method, and shooting method
- Calculate energy and approximate wave function in a `for` loop over initialized ℓ -values
 - `while` loop nested inside `for` loop for energy bounds

Numerov Method Function

```
def numerov(f, u, n, x, h):  
    nodes, c = 0, h**2/12.  
    f0, f1 = f(x), f(x+h)  
    for i in range(n):  
        x += h  
        f2 = f(x+h)  
        u.append(*insert Numerov equation here*)  
        f0, f1 = f1, f2  
        if u[-1]*u[-2] < 0.0:  
            nodes += 1  
    return u, nodes
```

Shooting Method Function

```
def shoot(En):  
    Global E  
    E, c, xm = En, (h**2)/6., xL + M*h  
    wfup, nup = numerov(f, [0,.1], M, xL, h)  
    wfdn, ndn = numerov(f, [0,.1], N-M, xR, -h)  
    dup = ((1+c*f(xm+h))*wfup[-1] -  
           (1+c*f(xm-h))*wfup[-3])/(h+h)  
    ddn = ((1+c*f(xm+h))*wfdn[-3] -  
           (1+c*f(xm-h))*wfdn[-1])/(h+h)  
    return dup*wfdn[-2] - wfup[-2]*ddn
```

Example Code

Bound States:

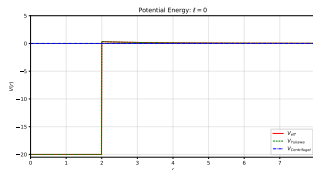
```
Lmax, Estart = 4, -V0
EL = []
for L in range(Lmax):
    while Estart < 0:
        E += dE
        .
        .
        .
        print(E, n, l)
    EL.append(E)
```

Pseudo-Bound States:

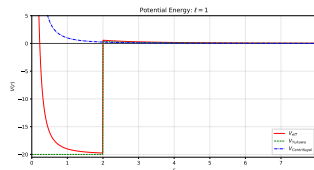
```
Lmax, Estart = 4, -V0
EL = []
for L in range(Lmax):
    while Estart <  $\exp(-\lambda*a)/a$ :
        E += dE
        .
        .
        .
        print(E, n, l)
    EL.append(E)
```

- 1 Background
- 2 Goals
- 3 Methods
- 4 Results & Discussion**
- 5 Moving Forward
- 6 AI Implementation
- 7 References

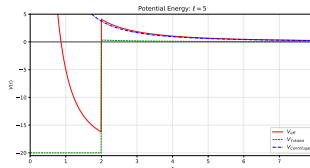
Potential Energy



(a) Potential energy V as a function of r for $\ell = 0$.

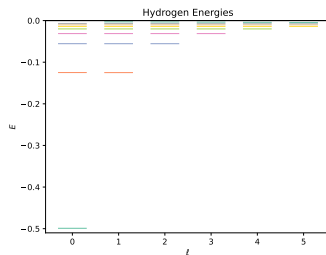


(b) Potential energy V as a function of r for $\ell = 1$.

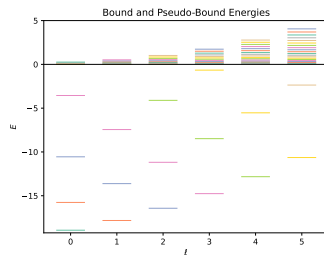


(c) Potential energy V as a function of r for $\ell = 5$.

Energy Plot

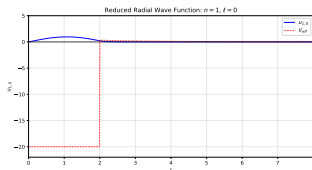


(a) Eigenenergies for Hydrogen up to $\ell = 5$.

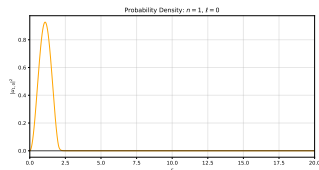


(b) Eigenenergies for Yukawa potential up to $\ell = 5$.

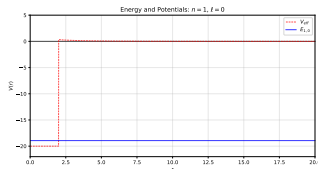
Wave Functions & Probability Densities



(a) Reduced radial wave function u_{10} as a function of r .

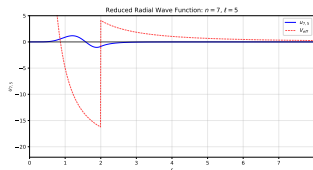


(b) Probability density for u_{10} as a function of r .

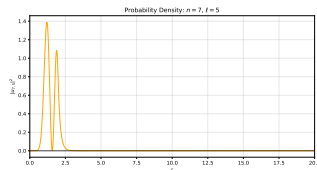


(c) Ground state energy E_{10} and potential as a function of r .

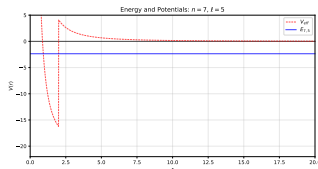
Wave Functions & Probability Densities



(a) Reduced radial wave function u_{75} as a function of r .

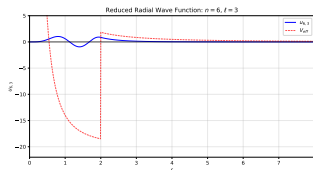


(b) Probability density for u_{75} as a function of r .

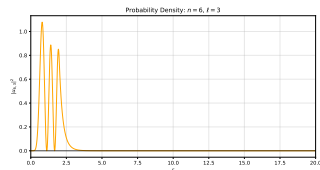


(c) Energy E_{75} and V_{eff} as a function of r .

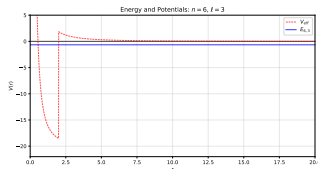
Wave Functions & Probability Densities



(a) Reduced radial wave function u_{63} as a function of r .

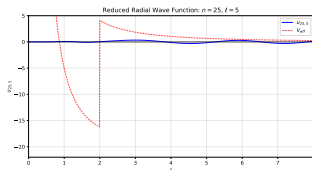


(b) Probability density for u_{63} as a function of r .

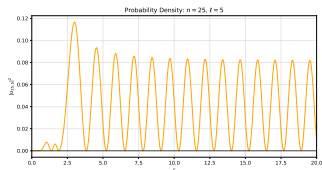


(c) Energy E_{63} and potential as a function of r .

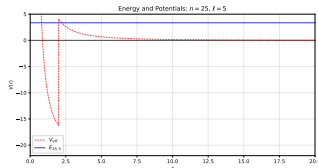
Wave Functions & Probability Densities



(a) Reduced radial wave function $u_{25,5}$ as a function of r .



(b) Probability density for $u_{25,5}$ as a function of r .



(c) Energy $E_{25,5}$ and potential as a function of r .

Discussion

- What does this data mean?

Discussion

- What does this data mean?
 - A computational model with the piecewise Yukawa potential is feasible

Discussion

- What does this data mean?
 - A computational model with the piecewise Yukawa potential is feasible
 - As a increases, more bound states appear

Discussion

- What does this data mean?
 - A computational model with the piecewise Yukawa potential is feasible
 - As a increases, more bound states appear
 - As V_0 increases, more bound states appear

Discussion

- What does this data mean?
 - A computational model with the piecewise Yukawa potential is feasible
 - As a increases, more bound states appear
 - As V_0 increases, more bound states appear
 - As λ increases, what happens?

- 1 Background
- 2 Goals
- 3 Methods
- 4 Results & Discussion
- 5 Moving Forward**
- 6 AI Implementation
- 7 References

Moving Forward

- Calculate tunneling probability for pseudo-bound and scattering states

Moving Forward

- Calculate tunneling probability for pseudo-bound and scattering states
- Determine at which well depth/width bound states start to appear

Moving Forward

- Calculate tunneling probability for pseudo-bound and scattering states
- Determine at which well depth/width bound states start to appear
- Continue to push the limits of λ to draw a more determined conclusion from its effects

- 1 Background
- 2 Goals
- 3 Methods
- 4 Results & Discussion
- 5 Moving Forward
- 6 AI Implementation**
- 7 References

AI Implementation

- Clarifying analysis of reference material

AI Implementation

- Clarifying analysis of reference material
- Code production, comparison, and debugging

AI Implementation

- Clarifying analysis of reference material
- Code production, comparison, and debugging
- Document formatting

- 1 Background
- 2 Goals
- 3 Methods
- 4 Results & Discussion
- 5 Moving Forward
- 6 AI Implementation
- 7 References

References

- [1] D. J. Griffiths, *Introduction to Quantum Mechanics*, 2nd ed. (Pearson Prentice Hall, 2005).
- [2] J. J. Wang, *Computational Modeling and Visualization of Physical Systems with Python*, (Wiley, 2016), Chap. 9.
- [3] H. Yukawa, "On the Interaction of Elementary Particles," *Proc. Phys.-Math. Soc. Japan* **17**, 48–57 (1935).
- [4] Yukawa Memorial, The University of Osaka, "About Hideki Yukawa," *The University of Osaka*. Available at: <https://www-yukawa.phys.sci.osaka-u.ac.jp/en/yukawahideki>.